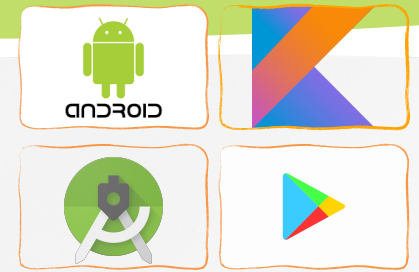


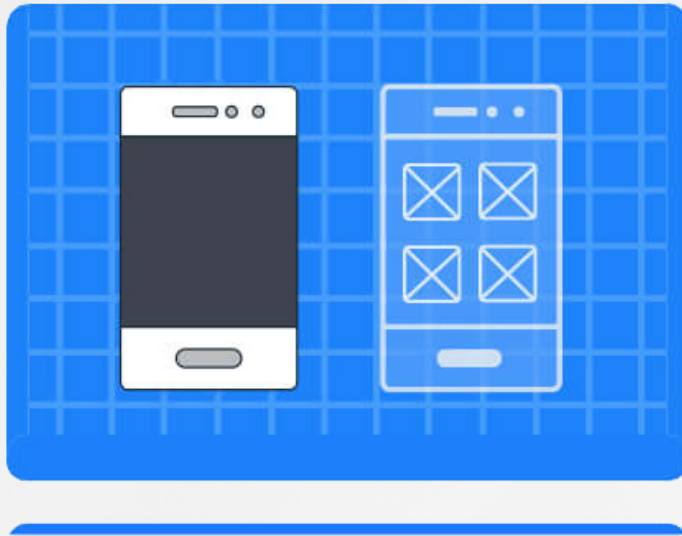
HorizontalScrollView

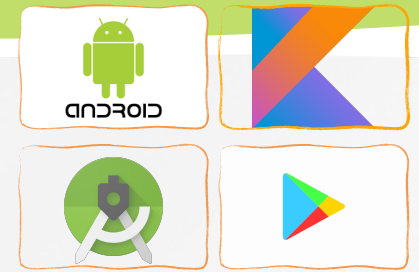
- This allow the user to graphically select an option from a set of small items (image, text...)
- The user interacts with the viewer using two simple actions:
 - Scroll the list (left ↔ right)
 - Click on a item to pick the option it offers.





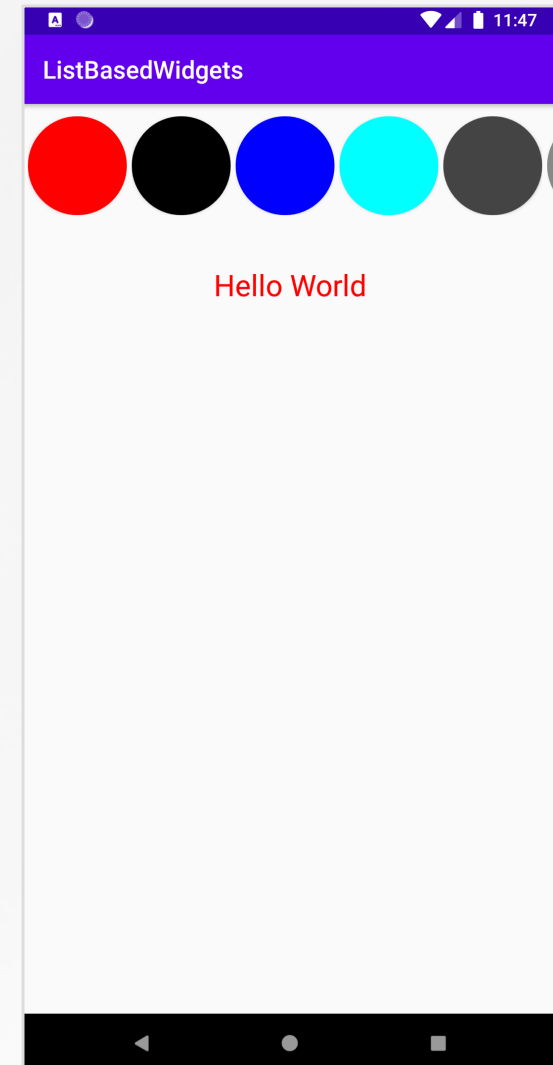
Example 08.7: Using the HorizontalScrollView

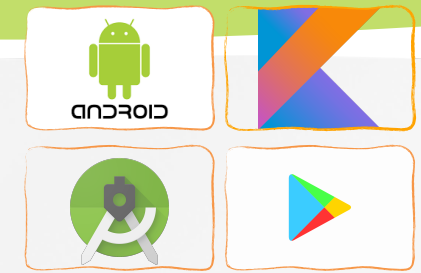




Using the HorizontalScrollView

- In this example we will place a HorizontalScrollView at the top of the screen, this view will show a set of color options.
 - The user may scroll through the thees color options and finally tap on an item.
 - As a result, the text color of TextView below the horizontal scroller will be changed





Using the HorizontalScrollView (cont.)

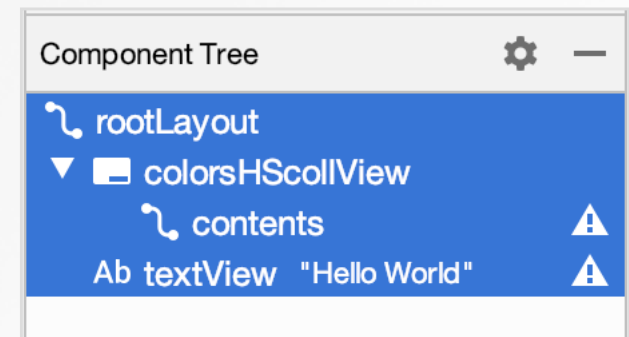
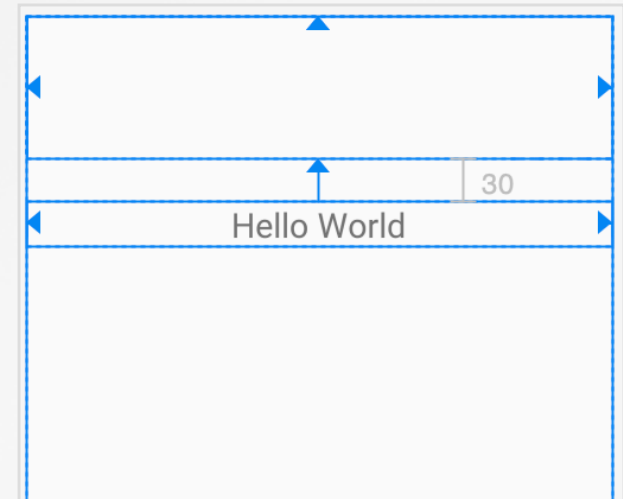
- Create the HorizontalScrollView within 'activity_main.xml'

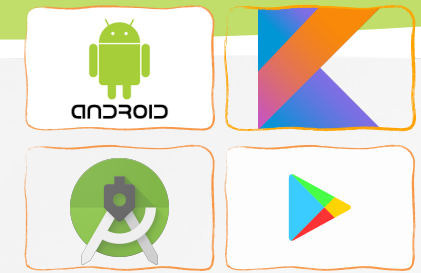
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://
schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" android:id="@+id/rootLayout">

    <HorizontalScrollView
        android:id="@+id/colorsHScrollView"
        android:layout_width="0dp"
        android:layout_height="100dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">
        <androidx.constraintlayout.widget.ConstraintLayout
            android:id="@+id/contents"
            android:layout_width="match_parent"
            android:layout_height="100dp">
            </androidx.constraintlayout.widget.ConstraintLayout>
        </HorizontalScrollView>

    ...
```

Horizontal
ScrollView





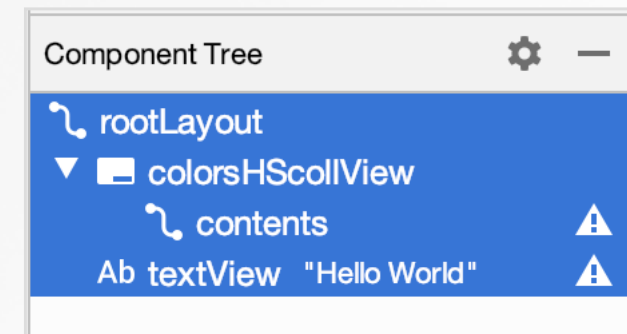
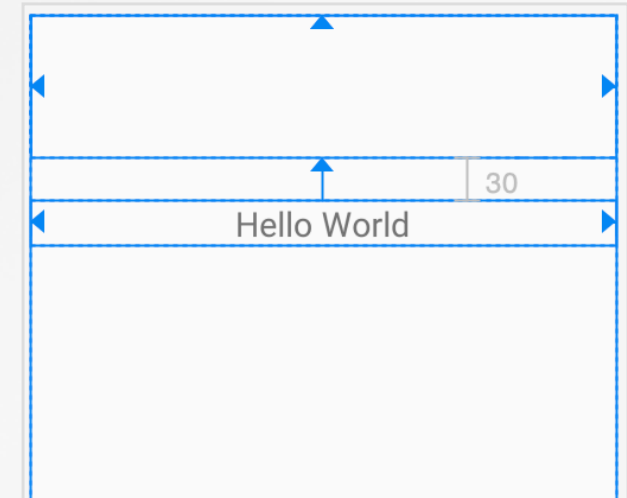
Using the HorizontalScrollView (cont.)

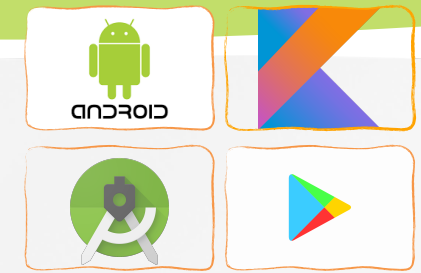
- Create the HorizontalScrollView within 'activity_main.xml'

...

```
<TextView
    android:id="@+id/textView"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="30dp"
    android:text="Hello World"
    android:textAlignment="center"
    android:textSize="24sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/colorsHScrollView" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

TextView with
'Hello World' text





Using the HorizontalScrollView (cont.)

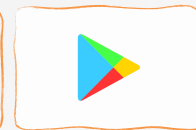
- Create the **item** within Layout (hd_list_item2.xml)

```
<?xml version="1.0" encoding="utf-8" ?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="100dp"
    android:layout_height="100dp">

    <Button
        android:id="@+id/colorBtn"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:background="@drawable/btn_background"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```





Using the `HorizontalScrollView` (cont.)

- Create the **background** for **button** within `Layout ()`

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <corners android:radius="40dp"/>
    <solid android:color="#6E6E6E"/>
</shape>
```

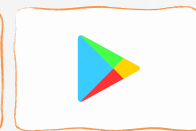




Using the HorizontalScrollView (cont.)

- Change our activity (*MainActivity.kt*)

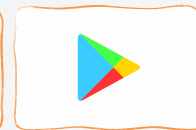
```
class MainActivity : AppCompatActivity() {  
    var colorsHScrollView : HorizontalScrollView? = null  
    var colors = arrayListOf<Int>()  
    var helloWorldTV : TextView? = null  
    var colorViews = arrayListOf<View>() //item views on scrollview  
  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContentView(R.layout.activity_main)  
  
        colorsHScrollView = findViewById(R.id.colorsHScrollView)  
        helloWorldTV = findViewById(R.id.helloWorldTV)  
  
        //Create color list  
        colors = arrayListOf( Color.RED, Color.BLACK, Color.BLUE, Color.CYAN, Color.DKGRAY,  
                               Color.GRAY, Color.GREEN, Color.MAGENTA, Color.YELLOW)  
  
        val content = addColorItemViews()  
        makeConstraintForContentView(content)  
    }  
    ...  
}
```

Using the HorizontalScrollView (cont.)

- Change our activity (*MainActivity.kt*)

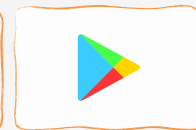
```
private fun makeConstraintForContentView(content: ConstraintLayout?) {  
    val constraintSet = ConstraintSet()  
    constraintSet.clone(content)  
    //The 1st color view item - parent  
    constraintSet.connect( colorViews[0].getId(), ConstraintSet.START,  
        ConstraintSet.PARENT_ID, ConstraintSet.START, 10);  
  
    for (i in 0..(colorViews.size - 2)) {  
        constraintSet.centerVertically(colorViews[i].id, ConstraintSet.PARENT_ID)  
        constraintSet.connect(colorViews[i].getId(), ConstraintSet.END, colorViews[i + 1].id,  
            ConstraintSet.START, 10);  
    }  
    //The final color view item - parent  
    constraintSet.centerVertically(colorViews[colorViews.size - 1].id, ConstraintSet.PARENT_ID)  
    constraintSet.connect(colorViews[colorViews.size - 1].getId(), ConstraintSet.END,  
        ConstraintSet.PARENT_ID, ConstraintSet.END, 10);  
  
    constraintSet.applyTo(content);  
}
```



Using the HorizontalScrollView (cont.)

- Change our activity (*MainActivity.kt*)

```
private fun addColorItemViews(): ConstraintLayout? {  
    val content = findViewById<ConstraintLayout>(R.id.contents)  
    content.setBackgroundColor(Color.LTGRAY) // background color for separating...  
  
    for (color in colors) {  
        //frame-0, frame-1, frame-2, ... and so on  
        val singleFrame: View = inflater.inflate(R.layout.hd_list_item2, null)  
        singleFrame.id = colors.indexOf(color)  
  
        val btn = singleFrame.findViewById<Button>(R.id.colorBtn)  
        btn.setOnClickListener {  
            helloWorldTV!!.setTextColor(color)  
        }  
  
        // set new background color  
        val background: GradientDrawable = btn.background.mutate() as GradientDrawable  
        background.setColor(color)  
        background.invalidateSelf()  
  
        colorViews.add(singleFrame)  
        content.addView(singleFrame)  
    }  
    return content  
}
```



Using the HorizontalScrollView (cont.)

- Run app

```
private fun addColorItemViews(): ConstraintLayout {  
    val content = findViewById<ConstraintLayout>(R.id.content)  
    content.setBackgroundColor(Color.LTGRAY)
```

```
    for (color in colors) {  
        //frame-0, frame-1, frame-2, ... and  
        val singleFrame: View = LayoutInflater.from(this).inflate(R.layout.frame, content, false)  
        singleFrame.id = colors.indexOf(color) * R.id.frame0
```

```
        val btn = singleFrame.findViewById<Button>(R.id.button)  
        btn.setOnClickListener {  
            helloWorldTV!!.setTextColor(color)  
        }  
    }
```

```
    // set new background color  
    val background: GradientDrawable = GradientDrawable()  
    background.setColor(color)  
    background.invalidateSelf()
```

```
    colorViews.add(singleFrame)  
    content.addView(singleFrame)  
}
```

